Cisco PE Node HSR-PE-07 QoS Congestion Event Resolution SOP

Node: HSR-PE-07 | **Vendor:** Cisco | **Severity:** Major | **Alarm Code:** ALARM_5367

Description

This SOP outlines the diagnostic and resolution procedures for mitigating QoS congestion events on Cisco PE nodes, specifically on the HSR-PE-07 node. The HSR-PE-07 node is a PE (Provider Edge) router located at the HSR Layout Access site, operating in a Cisco network environment. The failure domain in question is related to QoS (Quality of Service) and congestion events, characterized by microbursts, interface tail drops, and QoS queue starvation. This failure domain occurs due to the inherent limitations of the Cisco IOS-XE software architecture, where the QoS policy map configuration is processed in a hierarchical manner, with policies applied to individual classes of traffic. The Cisco Express Forwarding (CEF) algorithm, which is used for forwarding packets, can lead to congestion when the CEF table becomes saturated, resulting in microbursts and interface tail drops. Additionally, the Cisco QoS queueing mechanism can suffer from starvation when the queue is overloaded, leading to packet drops and QoS degradation. To mitigate these issues, it is essential to carefully configure and monitor QoS policies to ensure proper traffic handling and prevent congestion.

Pre-requisite Checks

- Verify that the HSR-PE-07 node is accessible and reachable via the network management system.
- Ensure that the redundant link is functioning correctly and is not causing the congestion event.
- Obtain the latest network topology and QoS policy configuration for the HSR-PE-07 node.

Potential Root Causes

Cause	Probability
Insufficient QoS policy configuration	High
CEF table saturation	Medium
QoS queue starvation	Low

Diagnostic Commands & Expected Outputs

```
Command: show policy-map interface GigabitEthernet0/0, Expected Output: Policy Map Configuration, Description: This command displays the QoS policy map configuration for the GigabitEthernet0/0 interface, allowing us to verify that the correct QoS policies are applied to the interface.

Command: show cef memory, Expected Output: CEF Memory Usage, Description: This command displays the CEF memory usage statistics, enabling us to determine if the CEF table is saturated.
```

```
Command: show queue interface GigabitEthernet0/0, Expected Output: Queue Statistics, Description: This command displays the queue statistics for the GigabitEthernet0/0 interface, allowing us to identify if QoS queue starvation is occurring.
```

Resolution Steps

Step 1: Step: Verify QoS policy configuration, Command: show running-config | begin policy-map, Description: Verify that the correct QoS policies are applied to the GigabitEthernet0/0 interface by checking the running configuration.

Step 2: Step: Adjust QoS policy configuration, Command: policy-map class bandwidth 100000, Description: Adjust the QoS policy configuration to increase the bandwidth for the specified class of traffic.

Step 3: Step: Clear CEF table, Command: clear cef, Description: Clear the CEF table to remove any stale entries and prevent CEF table saturation.

Step 4: Step: Verify QoS queue configuration, Command: show queue interface GigabitEthernet0/0, Description: Verify that the QoS queue configuration is not suffering from starvation by checking the queue statistics.

Post-Check Verification

```
Command: show policy-map interface GigabitEthernet0/0, Expected Output: Policy Map Configuration, Description: Verify that the QoS policy map configuration has been updated correctly by checking the policy map configuration for the GigabitEthernet0/0 interface.

Command: show cef memory, Expected Output: CEF Memory Usage, Description: Verify that the CEF memory usage statistics have improved by checking the CEF memory usage.

Command: show queue interface GigabitEthernet0/0, Expected Output: Queue Statistics, Description: Verify that the QoS queue statistics have improved by checking the queue statistics for the GigabitEthernet0/0 interface.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: copy running-config startup-config, Description: Restore the original QoS policy configuration by copying the running configuration to the startup configuration.

- Step: Clear CEF table, Command: clear cef, Description: Clear the CEF table to remove any stale entries and restore the original CEF table configuration.